

Project Name: **YVAM-480-US**

Unit Tag: **CH-1**

Qty.: **1**

Model: **YVAM1500**

Full Load - Design

PIN

YVAM1500A2	046BZHC7XB	SAXLXUSNX1	XXXXXXRM4M	2W1SXGA2B4	TC61010XXX	XSNSGXVXXX	XXXXXXEAS2	914C481XXX
....5...10	5...20	5...30	5...40	5...50	5...60	5...70	5...80	5...90
XXXXXXXXIXN	XMPG256377	1150370281	XXX60A0XX					
....5...100	5...110	5...120	5...130					

Unit

Model No.	YVAM1500
Number of Compressors	1
Compressor Type	Variable Speed
Number of Compressor Circuits	1
Capacity Control	10% - 100%
Refrigerant Type	R-1234ze
Compressor	M2C-233FAC
Cooling Mode Operation	
Cooling Capacity [kW]	1300
Total Input Power [kW]	424.2
Cooling [kW/kW]	3.064
NPLV.IP [kW/kW]	8.929
A-Weighted Sound Power [dB(A)]	97.5
Electrical Data	
Nominal Voltage / Voltage Limits	460/3/60 / 414V - 508V
Min. Circuit Ampacity [A]	696.0
Max. Fuse / CB Rating [A]	1000.0
Unit Short Circuit Withstand (STD) [kA]	100
Wires Per Phase	4
Wire Range (Lug Size)	4/0 - 500 kcmil
RLA [A]	571.5
Control kVA	5
Compressor kW	388.5



Performance Impacting Options

Fan Type	Container ship high efficiency fans with variable speed EC motor and harmonic filter
Weight & Dimensional Data	
Shipping Weight [kg]	10884
Operating Weight [kg]	11203
Refrigerant Charge [kg]	340
Overall Length [mm]	11887
Overall Width [mm]	2235
Overall Height [mm]	2503



Performance Report

Performance Specification

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Heat Exchanger Performance

Evaporator		Condenser (Air Cooled)	
Model	EA2914-481-CS-2	Ambient Air Temperature* [°C]	46.1
Entering Fluid Temperature* [°C]	25.00	Altitude* [m]	1000
Leaving Fluid Temperature* [°C]	17.00	Number of Fans	20
Flow Rate [L/s]	40.66	Total Air Flow [m3/s]	140
Fouling Factor* [m2 K/kW]	0.0176	Total Fan Power [kW]	35.79
Fluid Type* / Concentration* [%]	Propylene Glycol / 25.0		
Passes*	2		
Pressure Drop [kPa]	53.8		
Minimum Flow Rate [L/s]	24.24		
Maximum Flow Rate [L/s]	66.18		

* Designates user specified input

Certified in accordance with the AHRI Air-Cooled Water-Chilling Packages Certification Program, which is based on AHRI Standard 550/590(I-P) and AHRI Standard 551/591(SI). Certified units may be found in the AHRI Directory at www.ahridirectory.org. Auxiliary components included in total kW - chiller controls, heaters. Auxiliary power is already included in the compressor and fan power. Auxiliary components included in total kW - Chiller controls.



Part Load Performance (Based on Standard AHRI Unloading)

Percent Load	Ambient [°C]	Capacity [kW]	Power Input [kW]	Unit Efficiency [kW/kW]	Job FLA [A]
100.0	46.1	1300	424.2	3.064	571.5
75.0	33.6	974.9	189.6	5.143	109.5
50.0	21.1	649.9	65.69	9.896	93.5
25.0	12.8	325.0	17.06	19.04	90.5



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Sound Power Levels (In Accordance with AHRI 370)

Percent Load	Ambient [°C]	Octave Band Center Frequency [Hz]								LWA
		63	125	250	500	1000	2000	4000	8000	
100.0	46.1	103.5	96.5	96.5	93.5	92.0	87.5	89.5	87.5	97.5
75.0	33.6	100.5	94.0	94.0	91.5	89.5	86.5	86.0	81.0	95.0
50.0	21.1	93.5	87.5	87.5	85.5	83.5	81.5	81.0	75.5	89.5
25.0	12.8	83.5	79.5	77.0	78.5	74.5	77.5	76.5	88.0	88.5

Measurement of sound pressure used to obtain the sound power data presented is based on AHRI-370.

Air-cooled chillers are rated in terms of sound power not sound pressure. Johnson Controls provides estimates of sound pressure, but this is not the rating metric.

For an air-cooled chiller, sound pressure calculated from sound power varies depending on how the chiller is assumed to behave, i.e. the radiation model. In other words, determining sound pressure from sound power requires making assumptions that result in different answers at a given distance from the chiller. The environment also influences sound pressure in the field installation. Sound pressure estimation radiation models pertaining to air-cooled chillers include the 'traditional' hemispherical model, parallelepiped model and equivalent hemispherical model.

Regarding sound power, Johnson Controls references tolerance limits based on ASHRAE guidelines. These are +/- 6dB in the 63Hz octave band, +/- 4dB in all other octave bands and +/- 3dB for the overall dBA.

Tolerance limits are based on uncertainties associated with:

1. Measurement Test Procedure
2. Repeatability
3. Production / Manufacturing Variability

Standard deviation associated with air-cooled chiller sound data is a measure of spread i.e. it indicates the range of probability of sound levels. Note that for operating conditions other than AHRI's Standard Rating Condition, higher levels of uncertainty can be expected.

Lead times for factory performance testing depend on test laboratory availability. Please confirm with Johnson Controls Customer Service.

Notes:

Country of Origin: Mexico

Displacement Power Factor refers to compressor only. Unit Power Factor depends on fan option selected. Calculated value is available by request.

Minimum and maximum evaporator flow information are for full load ratings with Water.

The product image shown is for illustrative purposes only and is not representative of selected options.